

Software Engineer | AI Automation

Software Engineer with extensive experience in **Python (Django)**, and **AI Automation**. Proven track record of architecting scalable B2B marketplaces, industrial compliance platforms, and internal financial tools. Expert in integrating **LLMs** into production workflows and custom fine-tuned models on Hugging Face and optimizing business operations using **n8n** and **Streamlit**. Reduced hiring times by 30% and financial audit times by 23% through custom software solutions.

CORE SKILLS

Backend Engineering: Python, Django, Django REST framework, Golang (Gin), FastAPI, Celery, RabbitMQ, RESTful API
Infrastructure & DevOps: Docker, AWS (EC2), Amazon S3, Git/GitHub, DigitalOcean, Nginx, Kubernetes, CI/CD, Traefik, Azure App Services
AI & Machine Learning: n8n (Automation), Hugging Face, RAG, LangChain, Scikit Learn, Numpy, NLP, BERT, Fine Tuning, NLTK, pandas, Transformers, PyTorch
Database & Caching: PostgreSQL, MySQL, Redis, Vector Databases (FAISS, ChromaDB)
Frontend: HTMX, React.js, Next.js, JavaScript, Streamlit, Responsive Design
Tools & Integrations: Payment Gateways, Twilio, Google Analytics, Search Console, SEO, WebRTC, Webhooks, Postman, Jira, Trello
Languages English, Malayalam, Hindi, Tamil

WORK EXPERIENCE

ParkMedical Clinic, United Kingdom (Remote) Present
AI Automation Engineer

- Automated Financial Reconciliation:** Engineered a **N8N & Streamlit & PostgreSQL** system that automatically ingests and cross-verifies clinical schedules against billing ledgers. Replaced manual Excel matching with algorithmic checks, reducing discrepancy review time by **23%**.
- Intelligent Document Processing (n8n & AI):** Architected an automation pipeline that uses **LLMs** to classify incoming mail (HMRC, Legal/Court, Utility, Council) and detect urgency. Implemented smart routing logic that instantly alerts the respective teams via **Slack** for high-priority documents.
- Revenue Intelligence Bot:** Built a real-time Slack bot that queries live databases to report daily revenue streams and operational anomalies

Servtech Technical Services, United Arab Emirates Sep 2023 – Dec 2025
Software Developer, Dubai

- Talent Xchange (B2B Workforce Marketplace)**
B2B Architecture: Architected a multi-tenant platform for B2B workforce leasing. Integrated secure Payment Gateways (Stripe) to handle automated billing and subscription management. Event-Driven Performance: Engineered an asynchronous task queue using Celery and RabbitMQ to offload heavy background processes (bulk employee onboarding), ensuring zero latency for the core API.
- TalentPool (Recruitment Platform)**
AI Matching Engine: Built a recruitment platform achieving **158K+** impressions and a 2.6% CTR. Implemented Cosine Similarity algorithms (NumPy) to automate candidate scoring, reducing time-to-hire by **30%**.
Performance Engineering: Decoupled heavy matching algorithms from the main thread using Celery workers and RabbitMQ, ensuring zero latency for end-users during bulk data processing.

- **SEO & Performance:** Optimized the platform's core web vitals and SEO structure, driving a **400%** increase in organic visibility (100 to 500+ daily impressions).
- **CertTrace (Industrial Inspection Platform)**
 Digital Transformation: Developed an end-to-end compliance system for **Valve & Instrumentation** inspections, digitizing Job Orders, NODs (Notice of Deficiency), and field reporting.
 Dynamic PDF Generation: Integrated WeasyPrint to generate regulatory-grade Calibration and PM Certificates with complex "As-Found/As-Left" plotting

G-TEC Education Institute, IndiaOct 2021 - Nov 2022

Software Developer, Kerala

- Contributed to the development of an eCommerce web application using Python (Django) and React
- Gained hands-on experience building e-commerce components including payment integration, cart functionality, and order processing

Steyp EdTech company, IndiaMar 2021 - Sep 2021

Full Stack Developer, Kerala

- Developed full-stack components for food ordering system using Django REST Framework and React
- Created responsive UI components and RESTful APIs, gaining proficiency in production-level code standards

EDUCATION & CERTIFICATIONS

Swami Vivekananda Institute of Technology, India, Bachelor Degree in Computer Applications

Swami Vivekananda Institute of Technology, India, Higher Diploma in Information Technology

Steyp Computer Engineers & Tech Scientists, India, CS Engineering Course in Software Development

- Relevant Coursework: Python Programming, Django Framework, RESTful API Development, Database Management, Backend Development, Web Security
- Ecommerce Platforms, Software Deployment, Agile Methodologies, Industry Project Development

Certifications

Responsive Web Design, FreeCodeCampAug 2023

Scientific Computing with Python, FreeCodeCampAug 2023

Backend Developer, Steyp

Frontend Developer, Steyp

MCP : Automation in Production, Hugging FaceJun 2025

Generative AI, GoogleJun 2025

Extracurricular Activities

Google Cloud Innovator Badge, GoogleFeb 2024

Stack Overflow Contributions

Active participant in answering technical questions and providing solutions on the platform

GitHub Repository Contributions

Active contributor to various open-source projects on [Github](#)

Projects

Servtech Technical Services, TalentPool AI-Powered Recruiting Software, [Link](#)

Servtech Technical Services, Talent Xchange B2B Workforce Marketplace, [Link](#)

Servtech Technical Services, CertTrace – Valve & Instrumentation Inspection Platform, [Link](#)

AI Restaurant Call System, [Link](#)

- Natural language processing to understand customer intent and provide conversational responses
 - Designed and implemented an intelligent voice-based reservation system that enables restaurants to automate customer calls using advanced AI
 - Real-time speech recognition and text-to-speech for natural conversations
 - Fully automated reservation management with SMS confirmations
- The solution will significantly reduce staff workload by handling routine calls, improves customer experience through 24/7 availability

Microservices Authentication System (Golang + Kubernetes)

- Architected and deployed a production ready authentication microservice using Golang, PostgreSQL, and JWT based token authentication
- Implemented RESTful APIs with Gin framework, achieving stateless authentication for scalable distributed systems
- Orchestrated deployment on Kubernetes with 2 replicas, ConfigMaps, and health probes for high availability

Fine-Tuned and Deployed LLMs on Hugging Face, [Link](#)

- Fine-Tuning with Custom Data: Used the Unsloth to simplify the entire training process and handle custom data ingestion.
- Prepared and curated domain-specific datasets, then applied best practices to fine-tune Llama-based models (Llama-3.2-3b-instruct, DeepSeek-R1-Distill-Llama-8B) for improved accuracy and performance on specialized tasks.

Publications

AI Restaurant Call System, Medium

May 2025

Building an AI-Powered Restaurant Call System: A Deep Dive

Published technical article demonstrating the development of an intelligent voice-based reservation system using Python, FastAPI, LangChain, and CrewAI. The article provides a comprehensive breakdown of the system architecture, implementation challenges, and real-world business benefits of automating customer interactions for restaurants. [Link](#)